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NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Sixth Semester B.Tech (Minor in Computer Science) Degree Examinations**

May 2025

CS2009-1 – FUNDAMENTALS OF DATABASE MANAGEMENT SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following best describes a database?
A) A collection of unrelated data B) A structured collection of related data stored for retrieval
C) A set of tables without any relationships D) A file system
2. Which level of the three-schema architecture provides a user's view of the database?
A) Internal level B) Conceptual level
C) External level D) Physical level
3. What does a multi-valued attribute in ER model translate to in a relational model?
A) A single table column B) A separate table
C) A foreign key D) A derived attribute
4. Which type of attribute cannot be divided into smaller parts?
A) Composite attribute B) Derived attribute
C) Multivalued attribute D) Simple attribute
5. Which of the following statements is true about a primary key constraint?
A) It allows duplicate values but ensures that each value is unique within the table. B) It ensures that no two rows in a table can have the same value for the primary key attribute(s).
C) It can be null D) It can be created on any non-key column.
6. The operation which allows to process relationships from multiple relations rather than single relation is classified as
A) division operation B) relation operation
C) square operation D) join operation
7. Which of the following is a correct statement regarding a relational schema?
A) It defines the structure of data and the relationships between tables. B) It only stores data, not metadata
C) It refers to the actual data stored in the database. D) It cannot include any constraints
8. _____ is a special type of integrity constraint that relates two relations & maintains consistency across the relations
A) Entity Integrity Constraints B) Referential Integrity Constraints
C) Domain Integrity Constraints D) Domain Constraints

9. Which command is used to change the definition of a table in SQL?
 A) CREATE B) UPDATE
 C) ALTER D) SELECT
10. Drop Table cannot be used to drop a table referenced by a _____ constraint.
 A) Local Key B) Primary Key
 C) Composite Key D) Foreign Key
11. Which SQL statement is used to retrieve data from a database?
 A) Update B) Select
 C) Insert D) Delete
12. What is the result of the query **SELECT COUNT (DISTINCT Salary) FROM EMPLOYEE; ?**
 A) Total number of employees B) Total number of unique salaries
 C) Total number of salaries greater than a specified value D) Average salary of employees
13. Which of the following can activate a trigger?
 A) A user login event B) A SELECT query
 C) An INSERT, UPDATE, or DELETE operation D) A schema modification
14. You can delete a view with _____ command.
 A) drop view B) delete view
 C) remove view D) truncate view
15. Which operator is used to check the existence of a record in a subquery?
 A) In B) Exists
 C) Check D) For
16. What SQL statement would you use to create SQL database named Test?
 A) CREATE DBT Test; B) CREATE Test
 C) CREATE DATABASE Test; D) MAKE DATABASE Test;
17. Which of the following is a characteristic of functional dependency?
 A) It is a one-to-one mapping B) It requires foreign keys
 C) It is a constraint between two attributes D) It ensures primary key uniqueness
18. What is a minimal cover of a set of functional dependencies
 A) A set of dependencies with redundant attributes B) A minimal equivalent set of functional dependencies
 C) A set of all possible keys for a relation D) A reduced schema without any dependencies
19. A relation is in 1NF if it
 A) Has no transitive dependencies B) Has a candidate key
 C) Contains only atomic values D) Is free of all anomalies
20. Which of the following ensures the isolation property in concurrent transaction execution?
 A) Recovery algorithms B) Concurrency control mechanisms
 C) Deadlock detection mechanisms D) Buffer management

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe with a neat diagram of Three Schema Architecture.	6	L*2	1	1
	b) Design and construct an ER diagram for "Employee Management System". Analyze suitable structural constraints for the same. Also, for the same ER diagram show the 7- step ER to relational mapping.	10	L3	2	2
2.	a) What is DBMS? Discuss the main characteristics of database approach.	8	L2	1	1

- b) Explain the following terms with example
- | | | | | | |
|--|----------------------------------|---|----|---|---|
| i. Entity | ii. Attribute | 8 | L3 | 2 | 1 |
| iii. Referential integrity constraints | iv. Entity integrity constraints | 8 | L2 | 2 | 1 |
3. a) What is relation? Discuss the main characteristics of relations.
- b) Explain the following with examples
- | | | | | | |
|------------------------|--|---|----|---|---|
| i. Join operation | | 8 | L2 | 1 | 1 |
| ii. SELECT and PROJECT | | | | | |
- Unit – II**
4. a) Explain Aggregate functions with suitable examples
- b) Consider the following schema and apply the SQL commands for the queries given below:
- WORKS (Person_name, Company_name, Salary)
- LIVES (Person-name, Street, City)
- LOCATED-IN (Company_name, City)
- MANAGER (Person_name, Company_name)
- i. Find the names of all persons who live in 'KARKALA' city.
- ii. Retrieve the names of all persons of 'WIPRO' whose salary is between 10,000 and 25,000.
- iii. Find the names of persons who lives and works in the same city.
- iv. List the names of people who works for 'INFOSYS' along with the cities they live in.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 3 | 2 |
|--|--|---|----|---|---|
5. a) Define SQL. Explain the data types available for attribute used in SQL.
- b) Explain the following
- | | | | | | |
|----------------|--------------------------------|---|----|---|---|
| i. Drop | ii. Group by and Having clause | 8 | L2 | 3 | 1 |
| i ii. Order by | iv. IN Operator | 8 | L2 | 4 | 1 |
6. a) Write a creation of table and insert one tuple values for following database
- BRANCH (branch-name: string, branch-city: string, assets: real)
- ACCOUNT (accno: int, branch-name: string, balance: real)
- DEPOSITOR (customer-name: string, accno: int)
- CUSTOMER (customer-name: string, customer-street: string, customer-city: string)
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 4 | 2 |
|--|--|---|----|---|---|
- b) With an example, Explain the concept of View and trigger in SQL.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L2 | 4 | 1 |
|--|--|---|----|---|---|
- Unit – III**
7. a) Consider following relation and given functional dependencies:
- R = {A, B, C, D, E, F, G, H, I, J}**
- FD1: AB → C**
- FD2: A → DE**
- FD3: B → F**
- FD4: F → GH**
- FD5: D → I J**
- Identify the candidate key of R and Normalize the schema till maximum possible Normal Form.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 5 | 2 |
|--|--|---|----|---|---|
- b) What is read write transaction? Why concurrency control is needed?
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L2 | 5 | 1 |
|--|--|---|----|---|---|
8. a) Define minimal cover. Find the minimal cover of given FD's.
- E: {A → B, C → B, D → ABC AC → D}**
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 5 | 2 |
|--|--|---|----|---|---|
- b) What is normalization? Suggest and explain three different techniques to achieve 1NF with suitable example
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L2 | 5 | 1 |
|--|--|---|----|---|---|

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
